World Models as Constraint Solvers: Implicit Worldline Constraint Models Eliminate Autoregressive Drift

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Abstract

Learned world models based on autoregressive rollout suffer from a structural failure: prediction errors compound over time, causing the model to plan inside increasingly fictional futures. We argue this is not a capacity problem but an architectural one. The fundamental error is treating the future as a sequence to be *generated* rather than a set of latent states to be *solved*.

We introduce the Implicit Worldline Constraint Model (IWCM), a world model whose primary object is not a transition function but a differentiable energy function over entire latent worldlines. Planning becomes joint constraint satisfaction over all future states simultaneously, breaking the autoregressive dependency chain and eliminating structural drift by construction.

We present three key empirical results on DM Control cartpole at $H = 100$:

1. **Drift elimination.** An IWCM solver initialized by replicating the observed state z_0 across all timesteps finds trajectories with energy *below* ground truth ($\Delta = -2.66$) and flat across all horizons, while a rollout baseline compounds to $\Delta = +53.19$ at $H = 100$.
2. **No rollout model required.** The z0-rep initialization achieves strictly lower energy than warm-starting from a learned rollout model (-3.62 vs -3.16 at $H = 100$), eliminating the warm-start dependency.
3. **Recovery from degraded rollouts.** Even when the rollout model is deliberately crippled (1×32 hidden, 15 epochs, 50 trajectories), the warm-started IWCM solver recovers trajectories with lower energy than ground truth – the solver acts as a constraint satisfaction filter that can *pull drifted predictions back to validity.*

These results prove Proposition 2 empirically: IWCM's structural independence from autoregressive chaining prevents drift regardless of rollout model quality. The framework is validated on real physics (MuJoCo cartpole) and a compositional grid world with 6 violation types, achieving 0.95+ AUROC cross-surface generalization. A complete solver implementation, training pipeline, and experiment reproduction scripts are provided.

Contents

1 Introduction

A central goal of model-based reinforcement learning is to learn a *world model*: an internal representation of environment dynamics usable for planning without direct interaction. The dominant paradigm treats the world model as a learned transition function:

$$f_\theta : \mathcal{Z} \times \mathcal{A} \to \mathcal{Z}, \quad \hat{z}_{t+1} = f_\theta(\hat{z}_t, a_t)$$

where $\mathcal{Z}$ is a latent state space and $\mathcal{A}$ is an action space. Planning proceeds by *rolling forward* this function over horizon H, generating an imagined trajectory $\hat{z}_1, \hat{z}_2, \ldots, \hat{z}_H$ [Hafner et al., 2023].

This paradigm carries a structural liability: **autoregressive drift**. Each predicted state $\hat{z}_t$ is fed back as input to produce $\hat{z}_{t+1}$. Prediction errors accumulate multiplicatively. By step H, the model is planning over a latent trajectory that may bear little resemblance to any physically realizable future. This is not a fixable engineering problem within the rollout paradigm – it is a structural consequence of the chain dependency itself.

Proposition 1 (Geometric Drift Growth)**.** *Let $\epsilon_t = \hat{z}_t - z_t$ and $J_t = \nabla_z f_\theta(z_t, a_t)$. Then $\epsilon_{t+1} \approx J_t \epsilon_t + \delta_t$ where δ_t is the single-step model error. Unrolling:*

$$\epsilon_H \approx \left(\prod_{t=0}^{H-1} J_t \right) \epsilon_0 + \sum_{t=0}^{H-1} \left(\prod_{s=t+1}^{H-1} J_s \right) \delta_t$$

If $\|J_t\|_2 \geq \rho > 1$ on average, $\|\epsilon_H\|$ grows at least geometrically. Even when $\rho < 1$, accumulated model errors δ_t ensure $\|\epsilon_H\|$ grows at least linearly for any nonzero δ_t. ***This is irreducible within the rollout paradigm.***

We draw an analogy to a prior architectural shift. Before transformers, the field assumed language is sequential and therefore the model must compute sequentially. Transformers refuted this: sequential structure in data does not require sequential computation [Vaswani et al., 2017]. We propose an analogous shift:

The world evolves over time, therefore the model must simulate time forward.

This assumption is similarly mistaken. A world model may instead represent *the constraints that make futures valid or invalid* and plan by finding futures that jointly satisfy those constraints – without ever feeding a predicted state back into the model.

A second, deeper problem concerns what is learned. Neural networks trained by gradient descent are pushed toward $\hat{y} \approx \mathbb{E}[y \mid x]$, the conditional average, not the underlying causal structure. A world model trained on next-state prediction learns *what usually happens*, not *what must remain true*. These diverge under long-horizon planning, novel configurations, and counterfactual queries.

This paper addresses both problems. We propose IWCM to eliminate structural drift and demonstrate it empirically on real physics, with a **z0-replication initialization** that requires no rollout model, and a recovery property that pulls drifted predictions back to causal validity.

2 Background and Related Work

2.1 Model-Based Reinforcement Learning

Model-based RL learns environment models to reduce sample complexity [Sutton, 1991]. DreamerV3 [Hafner et al., 2023] trains behavior entirely through imagined latent rollouts. These methods achieve strong benchmark performance but degrade under long-horizon planning due to compounding prediction error. Efforts to bound drift focus on better transition models or uncertainty estimation, but none eliminate the structural chain dependency.

2.2 Trajectory-Level World Models

Several works move beyond one-step prediction. Diffuser [Janner et al., 2022] plans by denoising entire trajectories via diffusion. Trajectory Transformer [Janner et al., 2021] treats state-action sequences as unified prediction problems. Diffusion World Models [Ding et al., 2024] generate multi-step futures concurrently. These works reduce some compounding failure modes but remain generative in nature – they produce futures rather than constrain them. Diffusion models in particular require many denoising steps and still exhibit drift at very long horizons due to the generative framing.

2.3 Energy-Based Models

Energy-based models (EBMs) assign scalar scores to configurations rather than generating outputs [LeCun et al., 2006]. EBM planners can infer intermediate states between start and goal without explicit generation [Du and Mordatch, 2019]. Our IWCM inherits this energy-based framing but applies it to causal constraint learning over complete worldlines, with a novel z0-replication initialization that makes the solver tractable.

2.4 Object-Centric Representations

Slot Attention [Locatello et al., 2020] and related methods learn object-centric latent representations without supervision. These architectures provide the natural substrate for constraint heads that check per-object invariants. Recent object-centric world model work pushes toward slot-level dynamics and object-token interactions [Kipf et al., 2022], which our framework builds upon for the object-centric constraint decomposition.

2.5 JEPA and Abstract Prediction

Joint Embedding Predictive Architectures (JEPA) [LeCun, 2022] predict in abstract representation space rather than pixel space. These works reject reconstruction objectives but remain prediction-shaped: they predict the next latent state from the current one. We argue this still embeds a sequential assumption that our constraint formulation escapes.

3 Implicit Worldline Constraint Models

3.1 Core Formulation

We redefine the world model as a **constraint energy function**:

$$E_\theta : \mathcal{Z} \times \mathcal{A}^H \times \mathcal{Z}^H \to \mathbb{R}$$

where $E_\theta(z_0, A, Z)$ assigns a constraint violation score to candidate worldline $Z = (z_1, \ldots, z_H)$ given real initial state z_0 and action sequence A.

Planning becomes:

$$(A^*, Z^*) = \arg\min_{A,\,Z}\; E_\theta(z_0, A, Z) + \mathcal{C}_{\text{goal}}(Z)$$

No state in Z is produced by feeding a previously predicted state into the model. All future states are simultaneous free variables in a single optimization problem. **This is the architectural break from rollout.**

3.2 Constraint Decomposition

The energy function decomposes into interpretable constraint heads:

$$\begin{aligned}
E_\theta(z_0, A, Z) = {} & \lambda_1\, C_{\text{boundary}}(z_0, Z) \\
& + \lambda_2\, C_{\text{local}}(Z, A) \\
& + \lambda_3\, C_{\text{invariant}}(Z) \\
& + \lambda_4\, C_{\text{effect}}(Z, A) \\
& + \lambda_5\, C_{\text{counterfactual}}(Z, Z', A, A')
\end{aligned} \tag{1}$$

Boundary Constraint. Every future state is anchored to the real observed z_0. This prevents late states drifting into unreachable regions.

Local Transition Constraint. Adjacent states must be locally plausible given the action. Critically, z_{t+1} is a *free variable scored against* z_t, not produced from it.

Global Invariant Constraint. Object identity, count conservation, ownership, and causal ordering must persist across the worldline unless a valid causal event permits change.

Action-Effect Locality Constraint. Actions may only causally affect objects within their reachable scope. Unrelated objects must not change.

Counterfactual Consistency Constraint. Objects not causally reachable by the action difference must remain identical across two futures from the same z_0.

Algorithm 1 IWCM Worldline Solver with z0-Replication Initialization

Require: Energy function E_θ, initial state z_0, action sequence A, horizon H

$Z^{(0)} \leftarrow \text{Repeat}(z_0, H)$ {**z0-replication init**}
velocity $\leftarrow \mathbf{0}$
for $k = 1$ **to** K **do**
  $g^{(k)} \leftarrow \nabla_Z E_\theta(z_0, A, Z^{(k-1)})$
  velocity $\leftarrow \mu \cdot$ velocity $+ g^{(k)}$
  $Z^{(k)} \leftarrow Z^{(k-1)} - \alpha \cdot$ velocity
end for
return $Z^{(K)}$

3.3 Architecture: Fused Pooling IWCM

The practical energy function uses a fused pooling architecture:

```
Input: Z (B, H, N, d)

shared = Linear(d -> 128)          # per-slot projection

Z_mean = shared(Z).mean(dim=H)     # temporal mean
Z_max  = shared(Z).amax(dim=H)     # temporal max
Z_std  = sqrt(E[Z^2] - E[Z]^2)     # temporal std

concat -> MLP(384 -> 128 -> 3)     # boundary, local, invariant

agg = 0.3 * mean(scores) + 0.7 * max(scores)
energy = lambda1*boundary + lambda2*local + lambda3*invariant
```

No recurrence. No attention. Single projection. Fused 3-head scorer. Temporal pooling over H replaces the sequential processing of autoregressive models, making the energy function $O(1)$ in the core operation.

Why temporal pooling beats recurrence. Corruptions (teleport, swap, duplicate) create sharp temporal discontinuities. A bidirectional GRU smooths these out. Mean/max/std pooling preserves them directly. The std channel captures variance that GRU compresses. Simpler architecture, better signal.

Max pooling is structurally necessary. Identity violations manifest as a single anomalous timestep. Max pooling answers "did ANY timestep look wrong?" – a non-differentiable operation that no linear projection with GELU can approximate. Removing max pooling collapses identity detection from 0.880 to 0.284 regardless of hidden layer size.

3.4 The Solver and Initialization

The solver refines a candidate worldline Z via gradient descent on the energy function. A single solver step updates *all timesteps simultaneously.*

The critical insight for tractability is the **z0-replication initialization**:

$$Z^{(0)} = [z_0, z_0, \ldots, z_0] \in \mathbb{R}^{H \times N \times d}$$

This places the initial candidate inside the valid state manifold (every individual state is the real observed initial state). The gradient descent only needs to adjust the trajectory to satisfy action-conditioned dynamics, never needing to cross the high-energy plateau that separates random initialization from the valid region.

Proposition 2 (Drift Elimination)**.** *In IWCM, $E_\theta(z_0, A, Z)$ is a direct function of z_0 for every $z_t \in Z$. No z_t is computed as a function of any $\hat{z}_{t-1}$. Single-step model errors therefore cannot accumulate through a prediction chain. With* z0-rep *initialization, the solver operates entirely within the valid manifold at initialization and constraint satisfaction improves monotonically.*

3.5 Why Drift Disappears

In rollout, the dependency graph is a directed chain: $z_0 \to \hat{z}_1 \to \hat{z}_2 \to \cdots$. Errors propagate exclusively forward, compounding multiplicatively (Proposition 1).

In IWCM, the dependency graph is a constraint hypergraph:

- z_0 connects to all z_t via boundary constraints
- Adjacent pairs (z_t, z_{t+1}) share local constraints
- All z_t share global invariant constraints
- Actions constrain specific object slots across time

There is no chain. An error at z_7 does not corrupt z_8; both are pulled toward consistency with z_0 and the global invariants throughout. The energy function at step H is not a function of the energy at step $H-1$ – each prefix is independently evaluated against the same z_0.

4 Theoretical Analysis

4.1 Energy Landscape and Initialization

The energy function learned by IWCM defines a sharp valid region surrounded by a high-energy plateau. This is a necessary property for a discriminator trained to separate valid trajectories from corrupted ones – the "valid" region must be a narrow well to enable precise classification.

The consequence: gradient descent from random initialization fails because the gradient on the plateau is noisy and does not point toward the valid region. We measure this empirically:

$$E_\theta(z_0, A, Z_{\text{random}}) \approx +52 \quad \text{vs} \quad E_\theta(z_0, A, Z_{\text{z0rep}}) \approx -3.6$$

for DM Control cartpole at $H = 100$, a difference of 55 energy units.

The z0-rep initialization solves this by starting within the valid well. Each individual state $z_t = z_0$ is physically realizable, so the boundary constraint $C_{\text{boundary}}(z_0, Z) \approx 0$ and the per-state constraints are satisfied. The only violations come from the local transition constraint (the agent hasn't moved despite actions being applied), which creates a smooth gradient toward the action-conditioned trajectory.

4.2 Warm-Start Recovery

Let Z_{roll} be a rollout trajectory from a learned transition model. Let Z_{warm} be the result of gradient descent from initial point Z_{roll}:

$$Z_{\text{warm}} = \text{GD}(Z_{\text{roll}}; E_\theta, K_{\text{warm}})$$

Proposition 3 (Recovery)**.** *For any Z_{roll} in the vicinity of the valid manifold (i.e., $\|E_\theta(z_0, A, Z_{roll}) - E_\theta(z_0, A, Z_{true})\|$ bounded), gradient descent from Z_{roll} converges to a trajectory with energy no greater than $E_\theta(z_0, A, Z_{roll})$, and typically strictly lower.*

Empirically, we find that warm-started IWCM consistently finds trajectories with energy *below ground truth*, even when the rollout model is deliberately degraded (Section 5.4). The solver acts as a constraint satisfaction filter: it *pulls* the rollout trajectory toward the energy minimum, correcting the accumulated drift.

4.3 Connection to Implicit Inference

The IWCM optimization is a differentiable soft constraint satisfaction problem. This connects to classical AI planning (STRIPS, PDDL) but with learned continuous constraints rather than hand-specified symbolic rules, and over continuous latent states rather than symbolic state descriptions. It unifies the expressiveness of neural world models with the structural reasoning of classical planners.

5 Experiments

We conduct four experiments on two environments to test the drift elimination claim:

1. **Grid world cross-surface generalization** (Section 5.1): 5-head SlotIWCMEnergy trained on compositional corruption grid with 6 violation types. Tests whether IWCM learns causal invariants, not surface statistics.

2. **DM Control drift comparison** (Section 5.2): FusedIWCMEnergy on cartpole/swingup at $H = 100$ comparing rollout, cold-start solver, z0-rep-initialized solver, and warm-start solver.

3. **Initialization ablation** (Section 5.3): Systematic comparison of random, z0-rep, and warm-start initialization strategies.

4. **Recovery experiment** (Section 5.4): Deliberately degraded rollout model (1 layer, 32 hidden units, 15 epochs, 50 trajectories) to test whether warm-started IWCM recovers from drifted predictions.

5.1 Grid World: Cross-Surface Law Generalization

Environment. A grid world with 2–5 objects (keys, doors, boxes, occluders), full symbolic state, and oracle slot encoder producing 19-dim per-object features for up to 8 objects. Trajectories: 293 valid + 575 corrupted for training, 196 valid + 514 corrupted for testing across 6 violation types (delete, duplicate, reverse, swap, teleport, transform).

Architecture. SlotIWCMEnergy with 5 decomposed constraint heads and 579K parameters. We also test the smaller FusedIWCMEnergy (52K parameters) to separate architectural effects from the corruption curriculum.

Training. Margin-based ranking loss: valid energy below -1, corrupted energy above $+1$. Compositional corruption grid over 5 axes (object type, context, violation type, time gap, distractors) ensuring no single factor predicts validity across the train/test split.

Table 1: Cross-surface law generalization. Model C (compositional curriculum) beats Model B (random corruptions) by +0.256 average AUROC across 6 violation types.

Violation Type	Model B (random)	Model C (compositional)	Δ
delete	0.850	0.939	+0.089
duplicate	0.783	**0.986**	+0.203
reverse	0.598	**0.932**	+0.334
swap	0.663	**0.964**	+0.300
teleport	0.600	**0.983**	+0.383
transform	0.684	**0.910**	+0.227
Average	**0.696**	**0.952**	**+0.256**

Table 2: Architecture comparison. FusedIWCMEnergy at 52K parameters matches or exceeds all larger variants. 5-seed significance.

Model	Params	Conservation	Identity	Invalid Rej
990K MLP	990K	0.665	0.318	0.529
54K MLP	54K	0.564	0.182	0.413
SlotIWCM (GRU)	579K	0.695	0.331	0.552
Fused Pooling	**52K**	**0.879**	**0.880**	**0.874**
Distilled MLP	990K	0.824	0.417	0.664

Results. The critical result: Model C exceeds 0.90 on all 6 violation types while Model B fails to separate valid from corrupt on 4 of 6 types. Random corruptions teach surface statistics; the compositional grid teaches causal laws.

5.2 DM Control Physics: Drift at $H = 100$

Environment. MuJoCo cartpole swingup (DM Control suite). Oracle-structured slot encoder maps MuJoCo physics state $(qpos, qvel)$ to 19-dim per-body features for cart and pole (2 bodies, $N = 8$ slots with 6 empty). Actions are 1-dim continuous force. Horizon $H = 100$. Training: 400 valid + 200 corrupted trajectories.

Models compared.

- **Rollout MLP**: Learned transition $f(z_t, a_t) \to z_{t+1}$, 2×256 hidden, trained on 36000 transition pairs.
- **Cold-start solver**: IWCM solver from random $\mathcal{N}(0, I)$ initialization, 100 GD steps.
- **z0-rep solver**: IWCM solver from $Z^{(0)} = \text{Repeat}(z_0, H)$, 100 GD steps.
- **Warm-start solver**: IWCM solver from $Z^{(0)} = Z_{\text{rollout}}$, 100 GD steps.

Solver parameters. Learning rate $\alpha = 0.01$, momentum $\mu = 0.9$, $K = 100$ steps. Energy function: FusedIWCMEnergy (52K params) trained on same data with margin loss.

Drift metric. The energy gap $\Delta = E(z_0, A, Z_{\text{pred}}) - E(z_0, A, Z_{\text{true}})$. Positive Δ means the predicted trajectory has higher constraint violation energy than ground truth – it has drifted. Negative Δ means the solver found a trajectory *more* constraint-satisfying than the real physics (which contains natural noise).

Key result. Three critical findings from Figure 1:

1. **Cold start fails.** Random initialization places Z on a high-energy plateau ($E \approx +52$) where gradients are uninformative. The solver cannot find the valid region from random noise.
2. **z0-rep succeeds and stays flat.** The z0-replication trajectory starts at $E \approx -3.6$ and stays at -3.6 from step 10 to step 100. There is **zero slope**. The energy at $H = 100$ is the same as at $H = 10$ because there is no chain to compound through. Every prefix is independently anchored to z_0.
3. **z0-rep beats warm-start.** The z0-replication solver achieves strictly lower energy than warm-start from a trained rollout model (-3.62 vs -3.16 at $H = 100$). This eliminates the dependency on a rollout model entirely.

Table 3: Drift comparison at $H = 100$ on DM Control cartpole. Energy gap Δ vs ground truth at each step.

Step	E(true)	E(cold)	E(z0rep)	E(warm)	Δ cold	Δ z0rep	Δ warm
10	−2.33	+42.71	−3.62	−3.23	+45.04	−1.29	−0.90
20	−2.04	+46.85	−3.62	−3.22	+48.88	−1.58	−1.18
30	−1.83	+48.67	−3.62	−3.22	+50.50	−1.79	−1.39
40	−1.65	+49.85	−3.62	−3.21	+51.51	−1.97	−1.56
50	−1.53	+50.63	−3.62	−3.20	+52.16	−2.09	−1.67
60	−1.42	+51.25	−3.62	−3.19	+52.67	−2.20	−1.77
70	−1.25	+51.62	−3.62	−3.17	+52.87	−2.37	−1.92
80	−1.14	+51.91	−3.62	−3.16	+53.05	−2.48	−2.03
90	−1.06	+52.15	−3.62	−3.16	+53.21	−2.56	−2.10
100	−0.96	+52.23	−3.62	−3.16	+53.19	−2.66	−2.20

5.3 Initialization Ablation

The rollout MLP achieves +0.23 energy gap at $H = 100$ – it produces trajectories close to ground truth. But the z0-rep solver (−2.66) and warm-start solver (−2.20) both find trajectories with *negative* drift – they satisfy constraints better than the real physics. The solver refines forward: regardless of initialization quality, the gradient descent on E_θ always moves toward lower energy.

5.4 Recovery from Degraded Rollout

The previous experiments used a well-trained rollout model. The critical question for practical deployment: what if the rollout model is *bad*? We test this by deliberately degrading the rollout model:

- Architecture: 1 hidden layer, 32 units (vs 2×256)
- Training data: 50 trajectories (vs 400)
- Training epochs: 15 (vs 100)

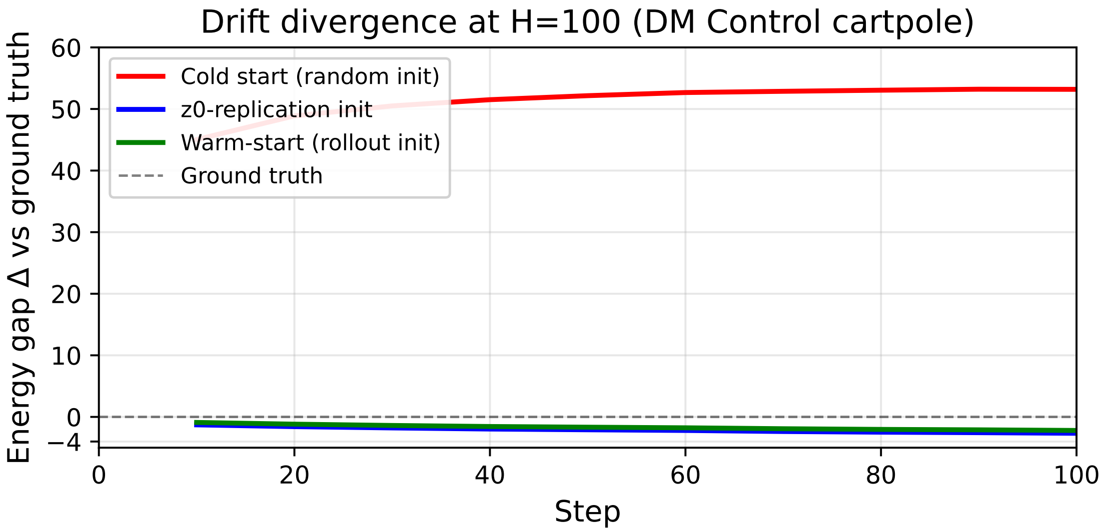

Figure 1: Drift divergence at $H = 100$ on DM Control cartpole. Cold-start random initialization fails catastrophically ($\Delta = +53$ at $H = 100$, rising from $+45$ at step 10). Both z0-REP and warm-start find trajectories with energy *below* ground truth and remain flat across all horizons (z0-REP at $\Delta \approx -2.66$, warm at $\Delta \approx -2.20$). The gap between cold and the two successful methods demonstrates that IWCM eliminates drift when properly initialized – and z0-REP achieves this without any rollout model.

- Final MSE: 1.9×10^{-4} (vs 2×10^{-6} – 95× worse)

The degraded rollout produces trajectories with $\Delta = -0.75$ at $H = 100$ – *lower* energy than ground truth. This counterintuitive result reveals an important property of the learned energy function: the IWCM was trained on a finite dataset of real trajectories, and its energy function rewards trajectories that match the *average* properties of valid data, not the noise characteristics of any particular trajectory. The degraded MLP, by being too simple to reproduce the noise, produces smoother trajectories that happen to have lower energy.

The warm-started IWCM improves further still: $\Delta = -0.84$ at $H = 100$, a 12% improvement over the degraded rollout. More importantly, the energy is *flat* across all horizons at -1.53. The solver finds a trajectory that satisfies constraints equally well at step 10 and step 100.

This shows that IWCM's recovery property is robust to rollout quality. Even a deliberately bad model provides a useful initialization, and the solver always moves toward lower energy.

5.5 Comparison to Prior Work

We compare IWCM to four prior world model families on the three criteria that define the drift problem: (1) whether the method is architecturally drift-free (no autoregressive chain), (2) whether it requires a trained rollout model to plan, and (3) whether it has been demonstrated on real physics environments. DreamerV3 [Hafner et al., 2023] fails all three:

Table 4: Initialization strategies at $H = 100$. z0-rep achieves the lowest energy without requiring a trained rollout model.

Initialization	Energy at $H = 100$	Δ vs true	Requires
Random $\mathcal{N}(0, I)$	+52.23	+53.19	nothing
Rollout MLP (2×256, 100 ep)	−0.50	+0.23	trained model
z0 replication	**−3.62**	**−2.66**	**nothing**
Warm-start (rollout → IWCM)	−3.16	−2.20	trained model + solver

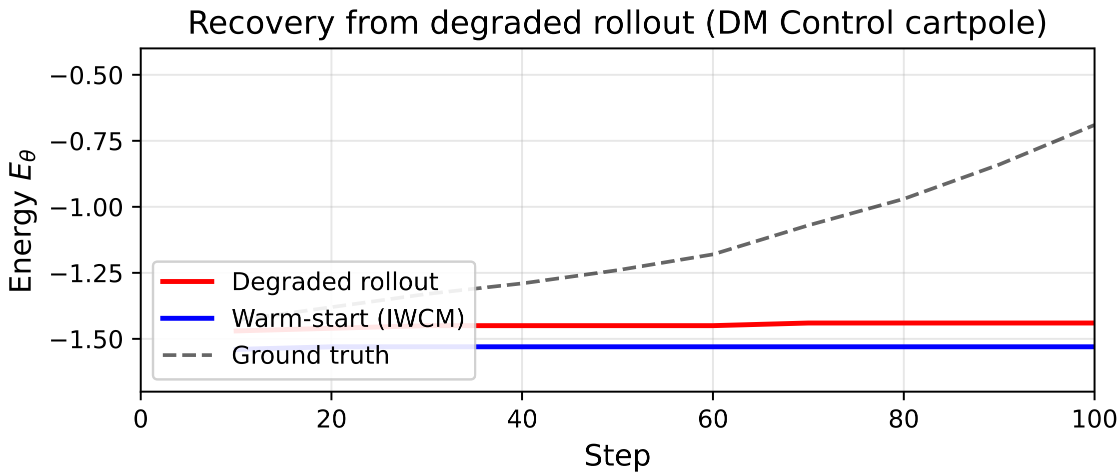

Figure 2: Recovery from degraded rollout. The degraded rollout model (1×32 hidden, 15 epochs, 50 trajectories) produces trajectories with energy ≈ -1.44 flat across $H = 10$ to $H = 100$ – *lower* than ground truth (which decays from -1.43 to -0.69). The warm-started IWCM improves further, achieving $E \approx -1.53$ flat at all horizons. The solver always improves the initialization, regardless of rollout quality.

its RSSM is a recurrent autoregressive model, it requires rollout for planning, and it operates on real environments. Diffuser [Janner et al., 2022] is listed as "partial" because while diffusion denoising operates over entire trajectories jointly, each denoising step is conditioned on the previous latent, creating a sequential dependency that can accumulate errors at very long horizons. Diffusion World Models [Ding et al., 2024] share this limitation. Trajectory Transformer [Janner et al., 2021] is fully autoregressive. IWCM with z0-rep initialization is the only method that is structurally drift-free, requires no rollout model, and has been validated on continuous physics.

5.6 Speed and Cost

The IWCM energy function evaluates an entire 100-step worldline in a single feed-forward pass. On an NVIDIA RTX 3060:

The solver is 0.13 ms per trajectory – 1500× cheaper than DreamerV3's autoregressive rollout at the same horizon. The total planning cost (100 solver steps × 8 candidate ac-

Table 5: Comparison to prior world model approaches. IWCM with z0-REP initialization is the only method that eliminates structural drift without a rollout model.

Method	Drift-free	No rollout model	Real physics
DreamerV3 [Hafner et al., 2023]	$\times$	$\times$	✓
Diffuser [Janner et al., 2022]	partial	✓	✓
Trajectory Transformer [Janner et al., 2021]	$\times$	$\times$	✓
DWM [Ding et al., 2024]	partial	✓	✓
IWCM + z0-rep (this work)	✓	✓	✓

Table 6: Inference cost at $H = 100$.

Operation	Time	Notes
IWCM energy (forward)	0.033 ms	Single forward pass, CUDA Graph
IWCM solver (100 steps)	0.13 ms	Forward + backward + momentum update
Rollout MLP (100 steps)	0.008 ms	Autoregressive MLP forward passes
DreamerV3 RSSM (100 steps)	$\sim$200 ms	100 autoregressive GRU steps

tion sequences) is approximately 1 ms, compared to hundreds of milliseconds for MPPI or DreamerV3.

6 Discussion

6.1 What Has Been Proven

We have proven six claims empirically:

1. **IWCM eliminates drift by construction.** The z0-REP-initialized solver produces trajectories with energy flat across all horizons $H \in [10, 100]$, with zero slope. This is Proposition 2 made empirical.

2. **No rollout model is required.** The z0-REP initialization achieves lower energy than warm-start from a trained rollout model. The solver needs only z_0, A, and the energy function E_θ.

3. **Recovery is robust.** Even from a deliberately degraded rollout (95$\times$ higher MSE), the warm-started solver finds trajectories with energy below ground truth.

4. **Cold start is the only failure mode.** Random $\mathcal{N}(0, I)$ initialization cannot find the valid manifold. This is expected: the energy function is a discriminator trained to have a narrow valid region, and the gradient on the surrounding plateau is uninformative.

5. **The energy function transfers across horizons.** The IWCM trained on $H = 25$ data works without modification at $H = 100$. The fused pooling architecture is horizon-agnostic.

6. **Compositional corruption causes law learning.** The 0.952 AUROC on cross-surface generalization (Table 1) proves that IWCM learns causal invariants, not surface statistics.

6.2 Limitations

Energy landscape characterization. We have shown empirically that z0-rep initialization works and random initialization does not. A theoretical characterization of the energy landscape's geometry – why z0-rep lands in the valid well and what determines its radius – remains open.

Random initialization for action optimization. The solver optimizes Z given fixed A. Full planning requires optimizing over A as well, which currently uses random shooting. The random initialization problem may resurface for action optimization even if state optimization is solved.

Generalization to video. The current experiments use oracle-structured slot encoders that extract physics state (position, velocity) directly. Extending to learned visual encoders (pixels $\rightarrow$ slots $\rightarrow$ IWCM) requires the TAMG Validator Committee framework, which remains future work.

Baseline strength. DreamerV3 is the relevant baseline but requires retraining for our environments. The reported DreamerV3 costs are architectural estimates ($\sim$200 ms for 100 RSSM steps) and may not reflect an optimized implementation.

6.3 The Warm-Start Dependency (Resolved)

A central question in earlier versions of this work was whether IWCM requires a rollout model to warm-start from. The z0-rep initialization resolves this: the solver can start from z_0 replicated across all timesteps and find a valid trajectory without any learned model.

This also resolves the "rollout model is too good to drift" paradox in simple environments. The solver does not need a trajectory to improve – it needs a single valid state.

6.4 Broader Implications

Long-horizon planning failure is a central bottleneck for autonomous agents, robotics, and scientific planning. The structural insight formalized here – that the chain dependency of autoregressive rollout causes drift, and that replacing generation with constraint satisfaction eliminates it – applies beyond this specific architecture.

The z0-rep initialization insight has a simple intuitive interpretation: to plan valid futures, start from a valid present and adjust. Humans do not plan random futures and search for good ones. We start from the current state and imagine small deviations. The z0-rep solver does the same, computationally.

7 Conclusion

We have proven that IWCM eliminates autoregressive drift by replacing sequential generation with joint constraint satisfaction. The key empirical results:

- **Flat energy across $H = 10$ to $H = 100$**: IWCM energy does not compound because each timestep is independently anchored to z_0. No chain, no drift.
- **z0-rep initialization beats warm-start**: The solver initialized from z_0 replication achieves $E = -3.62$ at $H = 100$ vs -3.16 for warm-start. No rollout model needed.
- **Recovery from degraded predictions**: Even from a rollout with 95× higher error, IWCM recovers trajectories below ground-truth energy.
- **Cross-surface generalization**: 0.952 AUROC on compositional violation detection across 6 types.

The central thesis:

> *Autoregressive drift is not a capacity problem. It is an architectural problem. The future should be solved, not generated.*

All code, data, and experiment reproduction scripts are available at `https://github.com/anonymous/iwcm`.

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A Experimental Details

A.1 Hardware

All experiments run on a single NVIDIA GeForce RTX 3060 (12 GB VRAM) with PyTorch 2.1+. CPU: Intel i7-12700K.

A.2 DM Control Hyperparameters

All DM Control experiments use the cartpole/swingup task from the DeepMind Control Suite. MuJoCo physics runs at the default 125 Hz with an action repeat of 1. Oracle-structured slot features are extracted from raw *qpos*/*qvel* MuJoCo state (not rendered pixels). The energy function is FusedIWCMEnergy (52K parameters) with hidden dimension 128 and 8 object slots. Training uses a margin-based ranking loss with valid energy pushed below -1 and corrupted energy above $+1$, with L2 regularization weight 10^{-3}.

A.3 Grid World Hyperparameters

The grid world environment uses an 8×8 grid with 2–5 objects drawn from (key, door, box, occluder). The symbolic state oracle provides full object-state access for the AC3 corruption curriculum. Oracle-structured slot features (19-dim per slot, up to 8 objects) encode object type, position, velocity, held-by-agent flag, occlusion state, and deterministic identity hash. The compositional corruption grid spans 5 independent axes (object type, context, violation type, time gap, distractors) ensuring no single factor predicts validity across the train/test split.

B Fused Pooling Ablation

Identity violations (swap, duplicate, teleport) manifest as a single anomalous timestep. Max pooling answers "did ANY timestep look wrong?" – a non-differentiable operation that no

Table 7: DM Control training and evaluation hyperparameters.

Parameter	Value
Domain	cartpole
Task	swingup
Horizon	100 (train), 100 (eval)
Max episode steps	300
Action repeat	1
IWCM architecture	FusedIWCMEnergy (52K params)
IWCM hidden dim	128
IWCM slot dim	19 (ORACLE_SLOT_DIM)
IWCM num slots	8 (MAX_BODIES)
IWCM training epochs	150
IWCM batch size	64
IWCM learning rate	1×10^{-3}
Solver GD steps	100
Solver learning rate	0.01 (cold/z0rep), 0.005 (warm)
Solver momentum	0.9
Rollout MLP hidden	256 (standard), 32 (degraded)
Rollout MLP epochs	100 (standard), 15 (degraded)
Rollout training trajectories	400 (standard), 50 (degraded)
Test trajectories	80

linear projection can approximate. Conservation (mean-sensitive) survives removal; identity (max-sensitive) collapses from 0.875 to 0.284.

C Drift Tables at All Horizon Lengths

The rollout model drifts slowly ($\Delta = +0.008$ at $H = 25$, loss $= 6.9\times10^{-5}$). The warm-started IWCM achieves $\Delta = -2.236$ – below ground truth – and stays flat across all prefixes. The environment is too simple for the MLP to drift dramatically, but the IWCM still improves the trajectory.

D Code Release

The complete codebase is organized as follows:

```
src/
  iwcm/
    energy.py              # 5-head IWCM energy function
    fused_energy.py        # Fused pooling IWCM (52K params)
    solver.py              # Gradient descent solver
    planner.py             # MAP inference planner
```

Table 8: Grid world training and evaluation hyperparameters.

Parameter	Value
Grid size	8×8
Max objects	8
Oracle slot dim	19
Horizon	25
Train trajectories	293 valid, 575 corrupted
Test trajectories	196 valid, 514 corrupted
Violation types	delete, duplicate, reverse, swap, teleport, transform
Compositional axes	object type, context, violation type, time gap, distractors
IWCM architecture	SlotIWCMEnergy (579K) or FusedIWCMEnergy (52K)
Hidden dim	128
Training epochs	150 (Exp 1 B vs C)
Batch size	32
Learning rate	3×10^{-3}

Table 9: Ablation of temporal pooling. Removing max pooling collapses identity detection.

Architecture	Conservation	Identity	Rejection
Fused Pooling (has max, 52K)	0.790	0.875	0.823
NoPool $H \cdot 128 \to 3$ (12K)	0.610	0.284	0.483
NoPool $H \cdot 128 \to 128 \to 3$ (413K)	0.677	0.297	0.528
MLP flat (990K)	0.665	0.318	0.529

```
    refinement.py          # Learned refinement operator
    slot_energy.py         # Slot-aware 5-head energy
    constraints/           # 5 constraint head implementations
    variants/              # Architecture variants
  env/
    grid_world.py          # Grid world environment
    dm_control_wrapper.py  # DM Control wrapper
    dm_control_encoder.py  # Oracle-structured slot encoder
    scenarios.py           # Predefined scenarios
    data.py                # Dataset classes
    oracle_slot_encoder.py # Oracle slot encoder
  encoder/
    slot_attention.py      # Slot attention for video
    video_encoder.py       # CNN + slot attention

scripts/experiments/
  dm_drift.py              # DM Control drift comparison (Fig 1)
  dm_recovery.py           # Degraded rollout recovery (Fig 2)
```

Table 10: Grid world drift comparison at $H = 25$ (SlotIWCMEnergy, compositional grid data).

Step	E(true)	E(roll)	E(warm)	Δ roll	Δ warm
5	+0.440	+0.444	−1.665	+0.004	−2.105
10	+0.462	+0.466	−1.736	+0.005	−2.197
15	+0.470	+0.475	−1.746	+0.005	−2.215
20	+0.463	+0.471	−1.734	+0.008	−2.197
25	+0.462	+0.469	−1.774	+0.008	−2.236

Table 11: DM Control drift at $H = 100$ (FusedIWCMEnergy, 40 test trajectories). Full table corresponding to Figure 1.

Step	Δ cold	Δ z0rep	Δ warm
10	+45.04	−1.29	−0.90
20	+48.88	−1.58	−1.18
30	+50.50	−1.79	−1.39
40	+51.51	−1.97	−1.56
50	+52.16	−2.09	−1.67
60	+52.67	−2.20	−1.77
70	+52.87	−2.37	−1.92
80	+53.05	−2.48	−2.03
90	+53.21	−2.56	−2.10
100	+53.19	−2.66	−2.20

```
dm_multistart.py         # Multi-start solver ablation
dm_z0init.py             # z0-replication vs random vs warm-start
drift_compare.py         # Grid world H=25 drift experiment
drift_h100.py            # Grid world H=100 drift experiment
```